

A Minimal Deployment Blueprint for Reader Privacy: Primary Cryptographic Path, Honest DHT Fallback, and Verifiable Receipts

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Abstract

This note is a capstone assembly: it states a minimal, referee-friendly deployment blueprint that composes the series’ contract objects into a single operational story. The blueprint is *tiered*: a primary, cryptographic path carries most queries; a DHT fallback handles residuals and is made *honest* by publishing explicit repetition/observation budgets and by recertifying when parameters drift. The note is intentionally non-redundant: definitions and interfaces are imported from their owners and this paper provides only the checklist glue.

1 Goal and scope

The series is primarily *provider- and auditor-facing*: it specifies how a deployment can make reader-privacy claims precise, checkable, and updateable. End-user verification is optional and minimal: we target *receipt discovery and anchoring validation* (not full behavioral attestation), via a small client-side interface.

A single theme motivates the blueprint: repeated use turns “one lookup privacy” into a *Boss Fight* budgeting problem. Dummy cover / equalization mechanisms can be formally sound yet practically inadequate under realistic query horizons Q ; therefore, the privacy boundary is architectural: push strong privacy into a small primary path, and treat the DHT as residual capacity whose leakage is explicitly budgeted and published.

Imports (owners). This note composes the full synthesis interface stack. Instead of re-enumerating every import in prose, we use the following grouping rule: core architecture and path semantics are owned by Synthesis 10, observation semantics by Synthesis 11 and Synthesis 15, publication/anchoring by Synthesis 9 and Synthesis 12, user discovery by Synthesis 13, and release/evaluation/change-control by the cited release, anonymity, and evaluation papers.[1, 2, 3, 4, 7, 8, 9, 10, 12, 13, 14, 15, 16]

2 Blueprint: what a deployment ships

2.1 Tiered paths

The architectural motivation, mixture bound, and honest-fallback semantics are owned by Synthesis 10 and should not be restated here.[7] This blueprint therefore treats the path design as an imported prerequisite:

- **Primary path:** a small cryptographic or strongly mediated surface with an explicit threat-window declaration and published receipt semantics.
- **Fallback path:** a DHT path whose repeated-use, observation, and path-selection costs are published honestly in the receipt chain.

What this note adds is the *operational packaging*: what artifacts, records, and role-specific checks a deployment should ship once it has adopted the tiered pattern.

2.2 Receipt chain: from build to public record

A deployment SHOULD ship a receipt chain with the following structure, each element expressed as a contract object [1].

1. **Manifest + canonicalization.** A stable manifest over source and configuration inputs, using the series' canonicalization/digest binding conventions (imported by the transparency interface [3]).
2. **ABOM/OINL declaration.** An ABOM/OINL object describing what is claimed and what is not claimed (owner: [13]).
3. **Receipt line items + replay hooks.** A receipt record carrying the minimal tuple in [4], including the declared TW and observation model (scalar or tiered). Each line item SHOULD include a `replay_hook` with `plan_id` and digest-bound `plan_spec_id`, so a third party can tell what replay workflow the receipt promises (and so drift triggers replay vs. recertification are well-defined).[5]
4. **Anchoring.** Append-only log anchoring, with inclusion (and, when relevant, consistency) proofs under the transparency interface [3].
5. **Optional UVI record.** A user-verifiable discovery/validation record, enabling clients to find the relevant receipt and validate anchoring mechanically [6].

Remark 1 (Non-goal: shipping all artifacts by default). *The archive's "source-only" policy is compatible with referee-grade defaults as long as the argument-bearing evidence is present in the papers and receipts. Scripts, datasets, and auxiliary tooling can remain on-request, but the published receipt chain must be sufficient to reproduce the stated bounds.*

3 Checklist by role (minimal MVP)

This section is the main export of the blueprint: a concise role-based checklist. It is phrased so a referee can audit it without reading the whole series.

3.1 Deployer checklist

1. **Declare the TW.** Publish a TW declaration object (threat window, adversary class, query budget conventions) [2].
2. **Choose tiering.** Specify the primary path (cryptographic) and fallback path (DHT) policy [7].
3. **Publish fallback honesty.** Decide whether fallback usage is secret-/state-independent; if not, publish the trigger model interface and selection-tax bound [9].

4. **Publish observation semantics.** Choose scalar p_{obs} or a tiered vector model and publish it in the receipt [10]. If a relay/stub-mix is used, record the attenuation assumptions [8].
5. **Budget repeated use.** Provide a Boss Fight dial-sheet slice or sufficient parameters for an auditor to reproduce the same bound [16].
6. **Emit the receipt chain.** Emit manifest, ABOM/OINL, receipt line items (including `plan_id/plan_spec_id`), and anchoring proofs [4, 12, 3, 5]. If any line item relies on schedule-only trace auditing, publish a plan certificate and log inclusion proof as part of the replay-plan evidence (Operational B).[11]
7. **Recertify on drift.** Bind receipts to the evaluation workflow and define recertification triggers (config changes, observed deviations, or policy changes) [14, 15].

3.2 Auditor/referee checklist

1. **Check anchoring.** Verify inclusion (and consistency when required) under the transparency interface [3].
2. **Check ownership.** Ensure claims are made only by the paper/object that owns the definition; imported semantics should cite their owners [1].
3. **Check TW match.** Confirm the receipt’s TW matches the paper claims and that the adversary/query-horizon conventions are coherent [2].
4. **Recompute bounds.** Verify each line item’s `plan_spec_id` resolves to a replay workflow you understand (Synthesis 18), then recompute the stated repeated-use budget from the published line items: Boss Fight tables [16], observation vector semantics [10], selection tax when applicable [9].
5. **Check change control.** Ensure the receipt chain is linked to release receipts (ABOM/OINL + property ledger) [12, 13] and that recertification triggers exist [15].

3.3 Client/user checklist (minimal, optional)

1. **Discover.** Obtain the UVI record (or equivalent discovery pointer) for the deployment [6].
2. **Validate anchoring.** Validate inclusion/consistency against published log checkpoints under the transparency interface [3].
3. **Interpret.** Read the published receipt tuple (TW, observation model, Q budget conventions) [4] and, if needed, consult the Boss Fight dial-sheet slice [16].
4. **Compare.** If the deployment publishes a watch policy or compare-profile adjunct, use it to compare the current claim surface against the last release without treating those adjuncts as new core receipt fields [17].

4 Minimal disclosure rules (to stay tight)

This note is “tight by construction” and therefore states what *must not* be duplicated:

- Do not restate the beacon/VRF/log interface outside Synthesis 9 [3]; cite it.

- Do not invent new receipt fields ad hoc; add them only via Synthesis 12 [4].
- Do not restate Boss Fight accounting in deployment papers; cite the dial sheet and publish parameters [16].
- Do not restate plan/state drift semantics; cite Synthesis 18 [5].
- Do not hide core evidence behind build flags (the series now uses explicit addenda when needed).

Exports

This note exports a single object: the role-based blueprint checklist and the claim that a deployment is referee-auditable when it publishes the receipt chain and supports the role checklists above. It does *not* export new math, new metrics, or new receipt primitives; those remain owned by the cited synthesis notes and series papers. For a concrete worked instantiation of the receipt chain, UVI record, and evaluation handoff, see the worked-example micro-note (Synthesis 17).[17]

References

- [1] Synthesis 0: *Contract Objects and Cross-Paper Interfaces*. (Unified archive.)
- [2] Synthesis 4: *Threat Windows for Anonymous DHT Claims*. (Unified archive.)
- [3] Synthesis 9: *Transparency Primitives Interface (Beacon/VRF/Log)*. (Unified archive.)
- [4] Synthesis 12: *Receipt Line-Item Schema*. (Unified archive.)
- [5] Synthesis 18: *Digest-Bound Replay Plans and State Declarations (Equivalence and Change Control)*. (Unified archive.)
- [6] Synthesis 13: *User-Verifiability Minimal Interface*. (Unified archive.)
- [7] Synthesis 10: *Tiered Deployment Pattern: Primary Cryptographic Path and Budgeted DHT Fallback*. (Unified archive.)
- [8] Synthesis 11: *Observation Attenuation via Relays/Stub-Mix*. (Unified archive.)
- [9] Synthesis 14: *Fallback Selection Tax (Secret-/State-Dependent Triggers)*. (Unified archive.)
- [10] Synthesis 15: *Tiered Observation Vectors*. (Unified archive.)
- [11] Operational B: *Auditable Trace Privacy with Abort: Disclosure Receipts for Side-Channel Resistant Transport*. (Unified archive.)
- [12] Release & Destination A: *Release Property Ledger Receipts*. (Unified archive.)
- [13] Anonymity C: *ABOM/OINL: An Anonymity Bill of Materials and Odds-Inflation Normal Form*. (Unified archive.)
- [14] Evaluation 1: *Notarized Anonymity Certificates*. (Unified archive.)
- [15] Evaluation 2: *Advantage Contracts and Stealth Audits*. (Unified archive.)
- [16] Boss Fight B: *AnonDHT Dial Sheet (Boss Fight Budgeting Tables)*. (Unified archive.)

- [17] Anonymous. Worked Example: Instantiating a Tiered Reader-Privacy Receipt. Synthesis 17, The-One-True-Archive (2026).
- [18] Anonymous. A Tiered Deployment Pattern for Reader Privacy: Primary Cryptographic Path, Budgeted DHT Fallback. Series synthesis note (Synthesis 10), 2026. (This archive.)